WHAT DOES A COMPUTER DO

- Fundamentally:
  - performs **calculations**

    a billion calculations per second!
  - **remembers** results

    100s of gigabytes of storage!
- What kinds of calculations?
  - **built-in** to the language
  - ones that **you define** as the programmer
- computers only know what you tell them

TYPES OF KNOWLEDGE

- **declarative knowledge** is **statements of fact**.
  - someone will win a Google Cardboard before class ends
- **imperative knowledge** is a **recipe** or "how-to".
  1) Students sign up for raffle
  2) Ana opens her IDE
  3) Ana chooses a random number between 1st and nth responder
  4) Ana finds the number in the responders sheet. Winner!

A NUMERICAL EXAMPLE

- square root of a number `x` is `y` such that `y*y = x`
- recipe for deducing square root of a number `x` (16)
  1) Start with a **guess**, `g`
  2) If `g*g` is **close enough** to `x`, stop and say `g` is the answer
  3) Otherwise make a **new guess** by averaging `g` and `x/g`
  4) Using the new guess, **repeat** process until close enough

g	g*g	x/g	(g+x/g)/2
3	9	16/3	4.17
4.17	17.36	3.837	4.0035
4.0035	16.0277	3.997	4.000002

WHAT IS A RECIPE

1) sequence of simple **steps**
2) **flow of control** process that specifies when each step is executed
3) a means of determining **when to stop**

1+2+3 = an **algorithm**!

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